Renal Reference — AKI Causes, Labs & Clinical Features

ETIOLOGY	CLINICAL FEATURES	LABORATORY FEATURES	COMMENTS
Prerenal azotemia	History of poor fluid intake or fluid loss (hemorrhage, diarrhea, vomiting, sequestration into extravascular space); NSAID/ACE-I/ARB; heart failure; evidence of volume depletion (tachycardia, absolute or postural hypotension, low jugular venous pressure, dry mucous membranes), decreased effective circulatory volume (cirrhosis, heart failure)	BUN/creatinine ratio above 20, FeNa <1%, hyaline casts in urine sediment, urine specific gravity >1.018, urine osmolality >500 mOsm/kg	Low FeNa, high specific gravity and osmolality may not be seen in the setting of CKD, diuretic use; BUN elevation out of proportion to creatinine may alternatively indicate upper GI bleed or increased catabolism. Response to restoration of hemodynamics is most diagnostic.
Sepsis-associated AKI	Sepsis, sepsis syndrome, or septic shock; overt hypotension not always seen in mild to moderate AKI	Positive culture from normally sterile body fluid or other test confirming infection; urine sediment often contains granular casts, renal tubular epithelial cell casts	FeNa may be low (<1%), particularly early in the course, but is usually >1% with osmolality <500 mOsm/kg
Ischemia-associated AKI	Systemic hypotension, often superimposed upon sepsis and/or reasons for limited renal reserve such as older age, CKD	Urine sediment often contains granular casts, renal tubular epithelial cell casts; FeNa typically >1%	
Nephrotoxin-Associated AKI: Endogenous			
Rhabdomyolysis	Traumatic crush injuries, seizures, immobilization	Elevated myoglobin, creatine kinase; urine heme positive with few red blood cells	FeNa may be low (<1%)
Hemolysis	Recent blood transfusion with transfusion reaction	Anemia, elevated LDH, low haptoglobin	FeNa may be low (<1%); evaluation for transfusion reaction
Tumor lysis	Recent chemotherapy	Hyperphosphatemia, hypocalcemia, hyperuricemia	
Multiple myeloma	Age >60 years, constitutional symptoms, bone pain	Monoclonal spike in urine or serum electrophoresis; elevated serum free light chains, low anion gap; anemia	Bone marrow or renal biopsy can be diagnostic
Nephrotoxin-Associated AKI: Exogenous			
Contrast nephropathy	Exposure to iodinated contrast	Characteristic course is rise in SCr within 1–2 d, peak within 3–5 d, recovery within 7 d	FeNa may be low (<1%)
Tubular injury	Aminoglycoside antibiotics, cisplatin, tenofovir, vancomycin, zoledronate, ethylene glycol, aristolochic acid, and melamine (to name a few)	Urine sediment often contains granular casts, renal tubular epithelial cell casts. FeNa typically >1%.	Can be oliguric or nonoliguric
Other Causes of Intrinsic AKI			
Glomerulonephritis/vasculitis	Variable (Chap. 326) features include skin rash, arthralgias, sinusitis (AGBM disease), lung hemorrhage (AGBM, ANCA, lupus), recent skin infection or pharyngitis (poststreptococcal), thrombotic microangiopathies including those related to drugs, such as cocaine, anti-VEGF agents, genetic abnormalities of the complement pathways	ANA, ANCA, Anti-GBM antibody, hepatitis serologies, cryoglobulins, blood culture, complement abnormalities, ASO titer (abnormalities of these tests depending on etiology)	Kidney biopsy may be necessary
Tubulointerstitial nephritis	Drugs are responsible for about 75% of the biopsy-proven acute interstitial nephritis, which involves tubules in most cases. Examples of causes include antibiotics, PPIs, immune checkpoint inhibitors. Non-drug-related causes include tubulointerstitial nephritis-uveitis (TINU) syndrome, lupus, viral infection (e.g., COVID, HIV, hantavirus), and *Legionella* infection.	Eosinophilia, sterile pyuria; often nonoliguric	Urine eosinophils have limited diagnostic accuracy; kidney biopsy may be necessary
TTP/HUS	Neurologic abnormalities and/or AKI; recent diarrheal illness; use of calcineurin inhibitors; pregnancy or postpartum; spontaneous	Schistocytes on peripheral blood smear, elevated LDH, anemia, thrombocytopenia	"Typical HUS" refers to AKI with a diarrheal prodrome, often due to Shiga toxin released from *Escherichia coli* or other bacteria; "atypical HUS" is due to inherited or acquired complement dysregulation. Diagnosis may involve screening for ADAMTS13 activity, Shiga toxin–producing *E. coli*, genetic evaluation of complement regulatory proteins, and kidney biopsy.
Atheroembolic disease	Recent manipulation of the aorta or other large vessels; may occur spontaneously or after anticoagulation; retinal plaques, palpable purpura, livedo reticularis, GI bleed	Hypocomplementemia, eosinophiluria (variable), variable amounts of proteinuria	Skin or kidney biopsy can be diagnostic
Postrenal AKI	History of kidney stones, prostate disease, obstructed bladder catheter, retroperitoneal or pelvic neoplasm	No specific findings other than AKI; may have pyuria or hematuria	Imaging with computed tomography or ultrasound

1. Prerenal azotemia

WHAT YOU SEE

Top row 'Prerenal azotemia'

WHAT IT ENCODES

Hypoperfusion of a structurally intact kidney (volume loss, HF, cirrhosis, sepsis, NSAIDs/ACE-I). The conserving nephron gives BUN:Cr >20, FENa <1%, urine Na <20, high specific gravity (>1.020), urine osm >500, and BLAND sediment (hyaline casts at most). It responds promptly to restoring perfusion — fully reversible before it tips into ischemic ATN.

BOARD TRAP

WRONG answer is calling a BUN:Cr >20 'always prerenal.' GI bleeding (urea from digested blood), high protein intake, steroids, and catabolic states also raise BUN disproportionately. Discriminator: confirm with FENa <1%, bland sediment, and clinical volume status — and remember a GI bleed elevates BUN by an extrarenal mechanism.

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2. Sepsis-associated AKI

WHAT YOU SEE

Row 'Sepsis-associated AKI'

WHAT IT ENCODES

Sepsis/hypotension causes AKI through hypoperfusion, inflammatory cytokines, and microvascular dysfunction. In the FIRST hours FENa can be low (<1%) with urine osm >500 before tubular necrosis develops — one of the classic low-FENa exceptions. As injury progresses it transitions to an ATN pattern (granular casts, FENa >1%).

BOARD TRAP

WRONG answer is using an early low FENa to exclude evolving ATN and withhold supportive care. Discriminator: in sepsis, low FENa is an EXCEPTION (early), not proof of pure prerenal physiology — track the urine sediment and creatinine trend; muddy-brown granular casts unmask the ATN.

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3. Ischemia-associated AKI

WHAT YOU SEE

Row 'Ischemia-associated AKI'

WHAT IT ENCODES

Sustained systemic hypotension (shock, major surgery, cardiac arrest), often with superimposed sepsis, produces ischemic ATN. Hallmarks: muddy-brown GRANULAR casts and renal tubular epithelial cell casts, FENa typically >2%, urine osm <350 (isosthenuric), BUN:Cr ~10-15. The proximal tubule and medullary thick ascending limb are most vulnerable.

BOARD TRAP

WRONG answer is giving repeated fluid boluses expecting prerenal-style improvement. Established ischemic ATN does NOT rapidly reverse with fluids — and overfilling causes pulmonary edema in an oliguric patient. Discriminator: muddy-brown casts + FENa >2% = ATN (supportive care, avoid further nephrotoxins); FENa <1% + bland urine = prerenal (volume responsive).

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4. Rhabdomyolysis

WHAT YOU SEE

Endogenous nephrotoxin row 'Rhabdomyolysis'

WHAT IT ENCODES

Muscle breakdown (crush injury, prolonged immobilization, seizures, statins, cocaine, prolonged exertion) releases myoglobin → pigment nephropathy. Labs: CK >5,000 (often >15,000), heme-positive dipstick with FEW/NO RBCs, hyperkalemia, hyperphosphatemia, HYPOcalcemia early, ↑uric acid. Treat with aggressive IV isotonic fluids targeting urine output 200-300 mL/h.

BOARD TRAP

Pigment nephropathy gives high FENa — this is the WRONG answer. Rhabdomyolysis (and other pigment/early sepsis/contrast states) often shows a LOW FENa (<1%) due to coexisting volume depletion and vasoconstriction. Discriminator: heme-positive dipstick without RBCs + CK >5,000 = myoglobinuric ATN; don't be fooled by the low FENa into calling it simple prerenal.

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5. Hemolysis

WHAT YOU SEE

Row 'Hemolysis'

WHAT IT ENCODES

Massive intravascular hemolysis (ABO-incompatible transfusion reaction, severe G6PD crisis, malaria, TMA) releases free hemoglobin → pigment-induced tubular injury. Labs: anemia, ↑LDH, ↓haptoglobin, ↑indirect bilirubin, hemoglobinuria (heme-positive dipstick, no RBCs), schistocytes if microangiopathic. Acute hemolytic transfusion reaction presents with fever, flank/back pain, and red urine during transfusion.

BOARD TRAP

WRONG answer is continuing a transfusion when the patient develops fever, flank pain, and dark urine. Discriminator: these signs = acute hemolytic transfusion reaction — STOP the transfusion immediately, give IV fluids, send a direct Coombs and recheck the crossmatch; do not 'push through' the reaction.

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6. Tumor lysis

WHAT YOU SEE

Row 'Tumor lysis'

WHAT IT ENCODES

Massive tumor-cell death (after chemo for bulky leukemia/lymphoma, or spontaneously) dumps intracellular contents: HYPERuricemia, HYPERkalemia, HYPERphosphatemia, and resulting HYPOcalcemia. Uric acid and calcium-phosphate crystals precipitate in tubules → crystal nephropathy/AKI. Prevent with IV hydration plus allopurinol (low risk) or rasburicase (high risk; contraindicated in G6PD deficiency).

BOARD TRAP

WRONG answer is giving rasburicase to a patient with G6PD deficiency. Rasburicase generates hydrogen peroxide and causes severe hemolysis/methemoglobinemia in G6PD-deficient patients. Discriminator: screen for G6PD before rasburicase; use allopurinol + aggressive hydration instead. Also: do NOT alkalinize urine in TLS — it worsens calcium-phosphate precipitation.

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7. Multiple myeloma

WHAT YOU SEE

Row 'Multiple myeloma'

WHAT IT ENCODES

Suspect in age >60 with bone pain, fatigue, anemia, hypercalcemia, and AKI. Mechanisms: cast nephropathy (Bence-Jones light chains obstruct tubules — the leading cause), hypercalcemia, and amyloidosis. Clues: monoclonal spike on SPEP/UPEP and serum free light chains, NARROW (low/normal) anion gap from cationic paraprotein, urine dipstick negative for protein despite heavy proteinuria (dipstick detects albumin, not light chains). Biopsy can be diagnostic.

BOARD TRAP

WRONG answer is excluding myeloma kidney because the urine dipstick is negative for protein. Standard dipstick detects ALBUMIN, not the filtered light chains of cast nephropathy. Discriminator: send UPEP / urine free light chains (and SPEP, serum free light chains) — a negative dipstick with a high urine protein:creatinine ratio points to light-chain proteinuria.

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8. Contrast nephropathy

WHAT YOU SEE

Exogenous nephrotoxin row 'Contrast nephropathy'

WHAT IT ENCODES

Iodinated contrast causes ATN via medullary vasoconstriction and direct tubular toxicity. Classic time course: Cr rises within 24-48h, peaks at 3-5 days, and recovers by 7-10 days; often NONoliguric. FENa is typically LOW (<1%) — a low-FENa exception. Risk factors: pre-existing CKD, diabetes, volume depletion, high contrast load. Prevention: isotonic IV saline.

BOARD TRAP

WRONG answer is attributing a creatinine rise 1-2 weeks after a procedure with aortic instrumentation to contrast. That timing and setting (livedo reticularis, blue toes, eosinophilia, low complement) point to ATHEROEMBOLIC disease instead. Discriminator: contrast nephropathy rises in 24-48h and recovers within a week; atheroembolic AKI is delayed, stepwise, and often irreversible.

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9. Tubular injury (drugs)

WHAT YOU SEE

Row 'Tubular injury': aminoglycosides, tenofovir, cisplatin, vancomycin, ethylene glycol

WHAT IT ENCODES

Direct tubular toxins → ATN: aminoglycosides (nonoliguric, ~day 5-7, dose/duration dependent, also Mg wasting), tenofovir and cisplatin (proximal tubulopathy/Fanconi), vancomycin (especially with concurrent piperacillin-tazobactam), ethylene glycol (calcium-oxalate crystals, high anion-gap + osmolar gap). Sediment: granular casts, tubular epithelial cells, FENa >1%.

BOARD TRAP

WRONG answer is dosing a nephrotoxin like vancomycin or an aminoglycoside by a single level or weight alone. Discriminator: these require therapeutic drug monitoring (trough/AUC), and the combination vancomycin + piperacillin-tazobactam markedly raises AKI risk — choose alternative beta-lactam coverage and monitor levels and daily creatinine.

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10. Glomerulonephritis / vasculitis

WHAT YOU SEE

Intrinsic row 'Glomerulonephritis/vasculitis'

WHAT IT ENCODES

Nephritic picture: hematuria with DYSMORPHIC RBCs, RBC CASTS, subnephrotic proteinuria, HTN, edema. Systemic clues: skin rash/palpable purpura, arthralgias, sinusitis (GPA), hemoptysis (anti-GBM/Goodpasture). Work up: ANA/anti-dsDNA, ANCA (PR3/MPO), anti-GBM, cryoglobulins, complement (low in lupus, post-strep, MPGN, cryo). RPGN with crescents needs urgent steroids ± cyclophosphamide/rituximab/plasmapheresis; biopsy often required.

BOARD TRAP

WRONG answer is waiting and rechecking labs in an RPGN with rising Cr and RBC casts. Crescentic GN can destroy kidneys within days. Discriminator: RBC casts + rapidly rising creatinine = medical emergency — get urgent nephrology, serologies, and biopsy and start empiric immunosuppression for life/organ-threatening disease (e.g., pulmonary-renal syndrome).

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11. Tubulointerstitial nephritis (AIN)

WHAT YOU SEE

Row 'Tubulointerstitial nephritis'

WHAT IT ENCODES

Immune/allergic interstitial inflammation, classically 5 days to weeks after a culprit drug (beta-lactams, sulfonamides, PPIs, NSAIDs, allopurinol, checkpoint inhibitors) or from infection/autoimmune disease. Findings: sterile pyuria, WBC casts, eosinophiluria, mild proteinuria, sometimes low-grade fever/rash. Treatment: STOP the offending drug; consider corticosteroids if no improvement.

BOARD TRAP

Always shows the classic fever-rash-eosinophilia triad — this is the WRONG answer. The full triad appears in fewer than 10-15% of AIN cases (and is least common with PPI/NSAID-induced AIN). Discriminator: don't exclude AIN because the triad is absent — a temporal drug link with sterile pyuria/WBC casts and rising Cr is enough to stop the drug; eosinophiluria is neither sensitive nor specific.

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12. TTP/HUS

WHAT YOU SEE

Row 'TTP/HUS'

WHAT IT ENCODES

Thrombotic microangiopathies. HUS (especially in children, Shiga-toxin E. coli O157:H7 after bloody diarrhea) = MAHA + thrombocytopenia + AKI. TTP (ADAMTS13 deficiency, adults) = MAHA + thrombocytopenia + neuro changes + fever + renal involvement (pentad, rarely complete). Smear: schistocytes; ↑LDH, ↓haptoglobin, ↑indirect bili, normal coags (distinguishes from DIC).

BOARD TRAP

WRONG answer is giving antibiotics for Shiga-toxin (STEC) HUS gastroenteritis or transfusing platelets in TTP. Antibiotics in STEC may increase toxin release and HUS risk; platelet transfusion in TTP can worsen microthrombi. Discriminator: STEC-HUS is supportive care (avoid abx/antimotility agents); TTP needs urgent plasma exchange — reserve platelets for life-threatening bleeding only.

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13. Atheroembolic / postrenal

WHAT YOU SEE

Bottom rows 'Atheroembolic AKI' and 'Postrenal AKI'

WHAT IT ENCODES

Atheroembolic (cholesterol embolization): days to weeks AFTER aortic instrumentation/angiography or anticoagulation — livedo reticularis, blue toes with intact pulses, retinal Hollenhorst plaques, EOSINOPHILIA, transient HYPOcomplementemia; stepwise, often irreversible. Postrenal: prostate/pelvic neoplasm, bilateral stones, retroperitoneal fibrosis → renal ultrasound/CT diagnostic; relieve obstruction (Foley, stent, nephrostomy).

BOARD TRAP

WRONG answer is anticoagulating an atheroembolic-disease patient or blaming the delayed AKI on contrast. Anticoagulation can PRECIPITATE further cholesterol embolization. Discriminator: blue toes + livedo + eosinophilia + low complement weeks after a vascular procedure = atheroembolic (stop anticoagulation, supportive care) — distinct from contrast nephropathy, which rises and recovers within a week.

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14. Lab features column

WHAT YOU SEE

Middle column 'Laboratory Features'

WHAT IT ENCODES

The column maps each etiology to its signature lab fingerprint — the matching exercise boards love: FENa (prerenal <1%, ATN >2%), sediment (bland=prerenal, muddy-brown granular=ATN, RBC casts=GN, WBC casts/eos=AIN, crystals=tumor lysis/ethylene glycol), and serologies (ANCA/anti-GBM/complement=GN, SPEP/UPEP=myeloma, CK=rhabdo, schistocytes=TMA).

BOARD TRAP

WRONG answer is treating FENa as a stand-alone diagnostic test. Multiple exceptions (sepsis early, contrast, rhabdo/pigment, GN, CKD, diuretic use) break the simple <1%=prerenal rule. Discriminator: always interpret FENa together with the urine SEDIMENT and clinical context — the casts are more specific than the FENa for localizing the lesion.
